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Value added immunoregulatory polysaccharides of *Hericium erinaceus* and their effect on the gut microbiota

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ABSTRACT

Hericium erinaceus polysaccharides (HEPs) were isolated from the fruiting bodies of *H. erinaceus* with 53.36 % total carbohydrates and 32.56 % uronic acid. To examine whether HEPs can alter the diversity and the abundance of gut microbiota, adult mice and middle-aged and old mice were fed with HEPs for 28 days. Based on the result of 16S sequencing of gut microbiota it was found that the relative abundances of *Lachnospiraceae* and *Akkermansiaceae* significantly increased, while the relative abundance of *Rikenellaceae* and *Bacteroidaceae* appeared to decrease. Bacterial solutions from different murine intestinal segments and feces were collected to ferment HEPs *in vitro*. It was found that HEPs remarkably promoted the production of NO, IL-6, IL-10, INF-γ and TNF-α. Moreover, HEPs significantly increased phosphorylation of signaling molecules, indicating that the immunomodulatory activity was completed *via* NF-κB, MAPK and PI3K/Akt pathways. Collectively, HEPs have potential to be developed as functional ingredients or foods to promote health.

1. Introduction

Hericium erinaceus, also known as "Houtou" in China and "Yamabushitake" in Japan, belongs to the family Hericiaceae, in the order Russulales. It widely grows in Asian countries. Several isolated compounds from *Hericium erinaceus* have shown numerous beneficial activities such as inhibition of ischemia injury-induced neuronal cell death (Lee et al., 2014), amelioration of Alzheimer's disease-related pathologies (Chiu et al., 2018), and promotion of catecholamine and nerve growth factor content in the central nervous system of rats (Shimbo, Kawagishi, & Yokogoshi, 2005), among others. Moreover, polysaccharides of the fruiting body of *Hericium erinaceum*, its liquid-cultured mycelia, and an alcohol-precipitated fraction of its culture filtrate showed antitumor activities (Mizuno, Saitô, Nishitoba, & Kawagishi, 1995), immunoregulatory effect (Yeh, Hsia, Chiu, Chiu, & Liu, 2011), besides abundant antioxidant properties (Yang, Lin, & Mau, 2002).

The human microbial community, including bacteria, viruses and protozoans, helps the host to protect against various external factors and hence offers a significant protection. The *Auricularia auricular* polysaccharides, amongst the fungal polysaccharides, were found to enrich the diversity and alter the composition of the fecal microbiota in mice with a significant decrease in the *Firmicutes/Bacteroidetes* ratio. The relative abundances of *Porphyromonadaceae* and *Bacteroidaceae* also were increased in the *A. auricular* polysaccharides fed mice, which was positively correlated with an increase in serum IgA and IgG concentrations (Zhao, Cheng, Nakata, Zhao, & Hu, 2019). A novel polysaccharide derived from *Flammulina velutipes* significantly decreased *Bacilli* at the class level compared with the control, and significantly reduced *Lactobacillales* and increased *Campylobacterales* at the order level compared with the control (Ye, Wang, Wang, & Deng, 2020). The *Lentinula edodes* polysaccharides evidently increased the Simpson index and the abundance of *Proteobacteria* at the phylum level after 4-week's oral administration compared with the control (Xu & Zhang, 2015). *Hericium erinaceus* polysaccharides, also belonging to the fungal polysaccharides, showed the potential effect of the modulation on the diversity and the abundance of gut microbiota.

In the past decade, many studies of the immunoregulatory effect of

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polysaccharides have been reported. For example, crude polysaccharides isolated from the fruit pulp of jackfruit enhanced the thymus weight index and the phagocytic rate after 30 days of subchronic p.o. administration at 4.5 mg/kg, and triggered immunostimulating activities in mice (Tan, Li, Lai, & Zhang, 2013). The buckwheat polysaccharide fraction from the seed of *Fagopyrum esculentum* showed immunoregulatory activities by increasing spleen and thymus indices and the leukocytes count in the blood of cyclophosphamide-induced immunosuppressed mice (Oh, Choi, Ha, Choi, & Park, 2018). A water-soluble β-D-glucan isolated from the fresh fruiting bodies of *Hericium erinaceus* induced the proliferation of lymphocytes and enhanced the expression of inflammatory cytokines IL-6, TNF-α and IL-1β secreted by THP-1 macrophages (Wu et al., 2019).

In our study, under the oral administration of *Hericium erinaceus* polysaccharides (HEPs), the diversity and the composition of the fecal microbiota in adult mice and middle-aged and old mice were investigated by 16S high-throughput sequencing. In addition, we collected bacteria solutions from intestinal different segments and feces of adult mice and middle-aged and old mice to ferment HEPs *in vitro*. A RAW 264.7/Caco-2 cells model was established to estimate the immunomodulatory effect of HEPs after the fermentation followed by molecular analysis.

2. Material and methods

2.1. Materials

The fruiting body of *Hericium erinaceus* polysaccharides were purchased from Jilin Jiaohe Songshan Food Co., Ltd. (Jiaohe City, Jilin Province, China). Cell Counting Kit-8 was obtained from APExBIO Technology LLC (Houston, USA). Neutral red staining solution, nitric oxide assay kit and BCA kit were obtained from Beyotime Biological Technology Co., Ltd. (Shanghai, China). Western Blot assay kit, SDS-PAGE kit, ELISA kits of TNF-α, IL-6, IL-10 and IFN were obtained from Boster Biological Technology Co., Ltd. (Wuhan, China). Anti-ERK, anti-phospho ERK, anti-JNK, anti-phospho JNK, anti-p38 MAPK, anti-phospho p38 MAPK, anti-Akt, anti-phospho Akt, anti-IkBα, anti-p65, anti-β-actin and anti-Histone H2A were purchased from ABclonal Biotechnology Co., Ltd (Wuhan, China). Inhibitors including BAY 11-7082, SB203580, SP600125, PD98059 and LY294002 were purchased from Selleck Chemicals (Shanghai, China).

2.2. Extraction of crude polysaccharides

Polysaccharides were isolated from the fruit body of the *Hericium erinaceus* according to Wang's method with some modifications (Wang, Luo, & Liang, 2004). Briefly, the dried *Hericium erinaceus* biomass was grounded into power and mixed with distilled water at a ratio of 1:10 w/v. The mixture was boiled for 8 h to extract the polysaccharides, and cooled to room temperature. Then, the mixture was filtered in vacuum by pump, and concentrated in a rotary evaporator under reduced pressure. The concentrated solution was added with a four-fold volume of ethanol to the concentrate overnight. The precipitate was collected by centrifugation, washed with deionized water three times.

The solution of the crude polysaccharides was treated with 5 % trichloroacetic acid for 30 min to remove residues proteins, and further centrifuged at 9 000×*g* for 30 min. The supernatant was concentrated in a rotary evaporator in vacuum, and then dialyzed in a 500–1000 D dialysis bag. The solution in the dialysis bag was collected and concentrated in a rotary evaporator in vacuum, and then lyophilized, affording HEPs.

2.3. Composition analysis

Total carbohydrates were quantified by the phenol-sulfuric acid method, using glucose as standard (Herbert, Phipps, & Strange, 1971). Reducing sugar content was evaluated by dinitrosalicylic acid (DNS) method (Miller, 1959). Uronic acid was quantified by the meta-hydroxydiphenyl method (Blumenkrantz & Asboe-Hansen, 1973).

2.4. In vitro fermentation of HEPs

2.4.1. Gut microbiota preparation

The fresh feces and intestinal contents (small intestine, cecum, colon) were collected from mice, and mixed and diluted with sterilized physiological saline to afford slurry (20 %, w/v). The slurry was filtered through four layers of gauze and then centrifuged at 500 rpm for 5 min to precipitate large insoluble particles. The resulting supernatants were collected and used for *in vitro* fermentation.

2.4.2. In vitro fermentation with gut microbiota

The basal medium was prepared as reported elsewhere (Díez-Municio, Kolida, Herrero, Rastall, & Moreno, 2016). The basic nutrient medium contained yeast extract (2.0 g/L), peptone (2.0 g/L), haemin (0.02 g/L), L-cysteine (0.5 g/L), bile salts (0.5 g/L), $NaHCO_3$ (2.0 g/L), NaCl (0.1 g/L), KH_2PO_4 (0.04 g/L), K_2HPO_4 (0.04 g/L), $CaCl_2 \cdot 6H_2O$ (0.01 g/L), $MgSO_4 \cdot 7H_2O$ (0.01 g/L), vitamin K_1 (0.01 g/L), resazur (0.01 g/L), and Tween-80 (2.0 mL/L). The pH was adjusted to 7.6. The fermentation medium was sterilized at 121 °C for 20 min. One mililiter suspension of feces, small intestinal content, fecal content, colonic content and 9.0 mL basal medium with 100 mg of HEPs were mixed in a sterilized Erlenmeyer flask. In a similar manner, mixtures of 1.0 mL of fecal suspension and 9.0 mL of basal medium with or without 100 mg of LPS were prepared as the experimental group and positive group, respectively. After fermentation for 0, 12, 24 and 36 h, all samples were centrifuged at 8000 rpm for 15 min. Then, the fermentation supernatants were filtered through a 0.22 μm membrane and then stored at −20 °C, prior to further use. Each experiment was repeated three times independently.

2.5. Immunomodulatory activity analysis

2.5.1. Cell lines and culture

Murine macrophage cell line (RAW264.7) and human intestinal epithelial cell line (Caco-2) were cultured in an DMEM medium supplemented with 10 % (v/v) fetal bovine serum and penicillin–streptomycin solution at 37 °C in a humidified incubator with 5 % CO_2. Cells were sub-cultured in the same medium under the same condition for further experiments.

2.5.2. Cytotoxicity assay by CCK-8

The RAW264.7 cells were seeded at a concentration of 1×10^5 cells/mL in 96-well plates. Cells were incubated with HEPs fermented by small intestine, cecum, colon and feces, LPS, without fermentation (CK). After 24 h, each well was added with 10μ L of CKK-8 solution and incubated for another 2 h. Absorbance was detected at 450 nm by a microplate ELISA reader.

The cell viability was calculated as follows:

$$\varphi = \frac{As - Ab}{Ac - Ab} \times 100, \%$$

where As is the absorbance value of sample group, Ab is the absorbance value of blank group, and Ac is that of control group (Liu, Song, Yang, Liu, & Zhang, 2007).

2.5.3. Nitric oxide production

The RAW264.7 cells (1×10^5 cells/well) were incubated with HEPs fermented by the small intestine, cecum, colon, feces, and LPS. In the control group (CK), the RAW264.7 cells were incubated with unfermented HEPs.After 24 h, 50 μL of the supernatants were isolated and mixed with an equivalent volume of Griess reagent. After 10 min, the

absorbance was measured at 540 nm using a microplate ELISA reader. Nitrite concentration was calculated from a $NaNO_2$ standard curve and was expressed as nmol per 1×10^5 cells.

2.5.4. Assay for phagocytic activity

The RAW264.7 cells (1×10^5 cells/well) were incubated in the culture medium in the absence (control) or presence of HEPs fermented. After 24 h, the cells were added with the neutral red dye and incubated for 1 h. The neutral red dye was removed. The cells were washed three times with the warm PBS and added with the acetate-ethanol cytolytic solution. After 2 h at 4 °C, the absorbance was detected at 540 nm using a microplate ELISA reader.

2.5.5. Cytokine detection in RAW264.7 cells or Caco-2/RAW264.7 co-culture system

The RAW264.7 cells were seeded at 1×10^5 cells per well in Transwell insert plates, and were cultured in a DMEM medium supplemented with 10 % (v/v) fetal bovine serum and penicillin–streptomycin solution at 37 °C in a humidified incubator with 5 % CO_2. At day 15, the Transwell insert plates were added by various concentrations of HEPs fermented by small intestine, cecum, colon, feces, and LPS. In the control group (CK), the RAW264.7 cells were incubated with unfermented HEPs. After 24 h, the cytokines in the basal medium were assayed according to the cytokine ELISA protocol from the manufacturer's instructions.

The Caco-2 cells were seeded at 1×10^5 cells per well in Transwell insert plates, and the cell cultures were the same as mentioned above. At day 15, RAW264.7 cells were seeded at 1×10^5 cells per well onto the basal side of the Transwell insert plates. After 24 h of coculture, the Transwell insert plates were added by various concentrations of fermented HEPs by small intestine, cecum, colon, feces, and LPS. In the control group (CK), the RAW264.7 cells were incubated with unfermented HEPs. After co-culturing for additional 24 h, the cytokines in the basal medium were assayed according to the cytokine ELISA protocol from the manufacturer's instructions (Xu, Yang, Ning, & Zhang, 2015).

2.5.6. Western blot analyses

The RAW264.7 cells were seeded in 6-well culture plates at a concentration of 1×10^6 cells/mL for 24 h and were then treated with or without specific inhibitors (NF-κB pathway inhibitor 10 μM BAY 11-7082, MAPK pathway inhibitor 20 μM SB203580, 10 μM SP600125, 20 μM PD98059, or Akt pathway inhibitor 20 μM LY294002) for 1 h. After 1 h, cells were treated for 3 h with various concentrations of HEPs. Next, the cells were washed three times with ice-cold PBS and centrifuged at 2000×g for 3 min. And the supernatant was removed. Cells were lysed with RIPA for 1 min on ice, and the cell lysates were centrifuged for 10 min (9000×g) at 4 °C. The proteins in the supernatant were determined using the BCA assay. The proteins (40 μg) were separated by electrophoresis of a 10 % sodium dodecyl sulfate-polyacrylamide gel (SDS-PAGE) and transferred onto a PVDF membrane. The membrane was blocked with 5 % skim milk in TBST for 1.5 h and then detected by the ECL Western blot detection kit to expose signals to X-ray films. Protein expression was analyzed using the software Quantity one (version 4.6, Bio-Rad, USA) (Kang et al., 2011).

Table 1
The effect of HEPs treatments on body weight.

Group	Initial weight (g)	Final weight after experiment (g)	Weight gain percentage (%)
Y	29.31 ± 0.608	44.07 ± 1.954	50.36 %
YL	30.54 ± 0.607	43.44 ± 11.684	42.24 %
YH	30.80 ± 0.650	43.16 ± 3.476	40.13 %
N	41.84 ± 1.049	46.66 ± 3.003	11.52 %
NL	40.73 ± 0.700	45.97 ± 2.593	12.87 %
NH	41.70 ± 1.116	46.66 ± 2.981	11.89 %

Y, 8-week-old mice gavaged at the same volume of sterile physiological saline. YL, 8-week-old mice gavaged at low dose (500 mg /kg body weight). YH, 8-week-old mice gavaged at high dose (1000 mg /kg body weight). N, 6-month-old mice gavaged at the same volume of sterile physiological saline. NL, 6-month-old mice gavaged at low dose (500 mg /kg body weight). NH, 6-month-old mice gavaged at high dose (1000 mg /kg body weight). Data are expressed as mean ± SD (n = 7).

2.6. Animals experiments

2.6.1. Experimental design

All experiments were performed in compliance with the relevant laws and were conducted at the Laboratory Animal Center of Jilin University of College of Animal Science. Specific pathogen-free male Kunming mice (8-week-old and 6-month-old) were acclimatized for at least 14 days in a temperature of 22 °C ± 2 °C with a 12 h light-dark cycle with free access to water and standard rodent chow. The mice were grouped as follows: (1) 8-week-old mice were randomly divided into three groups (control group, low dose group and high dose group, n = 7). In the low dose group, the mice were gavaged of the HEPs at a daily dose of 500 mg /kg body weight (b.w.). In the high dose group, the mice were gavaged of the HEPs at a daily dose of 1000 mg /kg b.w.. In the control group, the mice were gavaged at the same volume of sterile physiological saline. (2) 6-month-old mice were randomly divided into three groups (control group, low dose group and high dose group, n = 7). In low dose group, the mice were gavaged of the HEPs at a daily dose of 500 mg/kg b.w.. In the high dose group, the mice were gavaged of the HEPs at a daily dose of 1000 mg/kg b.w.. In the control group, the mice were gavaged at the same volume of sterile physiological saline. These mice were treated for 28 consecutive days with body weight monitored daily. On the last day, mice were sacrificed by cervical dislocation. The feces, small intestine, cecum and colon tissues were collected separately and immediately stored at −80 °C for further analysis.

2.6.2. DNA extraction and high throughput sequencing

Genomic DNA was extracted following the manufacturer's protocol. The V4 hypervariable region of the 16S ribosomal RNA (rRNA) gene from DNA was applied for amplification. The high throughput sequencing was completed by Beijing Novogene Technology Co., Ltd (Beijing, China). After the preliminary quality filtering, the entire qualified DNA was used to construct a library. The jagged ends of DNA fragment were converted into blunt ends by T4 DNA polymerase, Klenow fragment and T4 polynucleotide kinase. To add adapters easily, an 'A' base was added to each 3`end. For genomic DNA, a fusion primer with dual index and adapters was used for PCR. Fragments that were too short were removed by Ampure beads. The qualified library was used for sequencing with the bioinformatics analysis. And the raw data were eliminated the adapter pollution and low quality to obtain clean reads and then paired-end reads with overlap were merged to tags. Tags were clustered to operational taxonomic unit (OTU) at 97 % sequence similarity. Taxonomic ranks were assigned to OUTs representative sequence by Ribosomal Database Project (RDP) Na, e Bayesian Classifier v.2.2. Finally, alpha diversity, beta diversity and different species screening were analyzed based on OTUs and taxonomic ranks.

2.7. Statistical analysis

Data were presented as the mean ± standard deviation (SD). The significance of difference was evaluated with one-way ANOVA, followed by Post Hoc in Statistical Analysis software SPSS 23. Significant differences were set at $P < 0.05$.

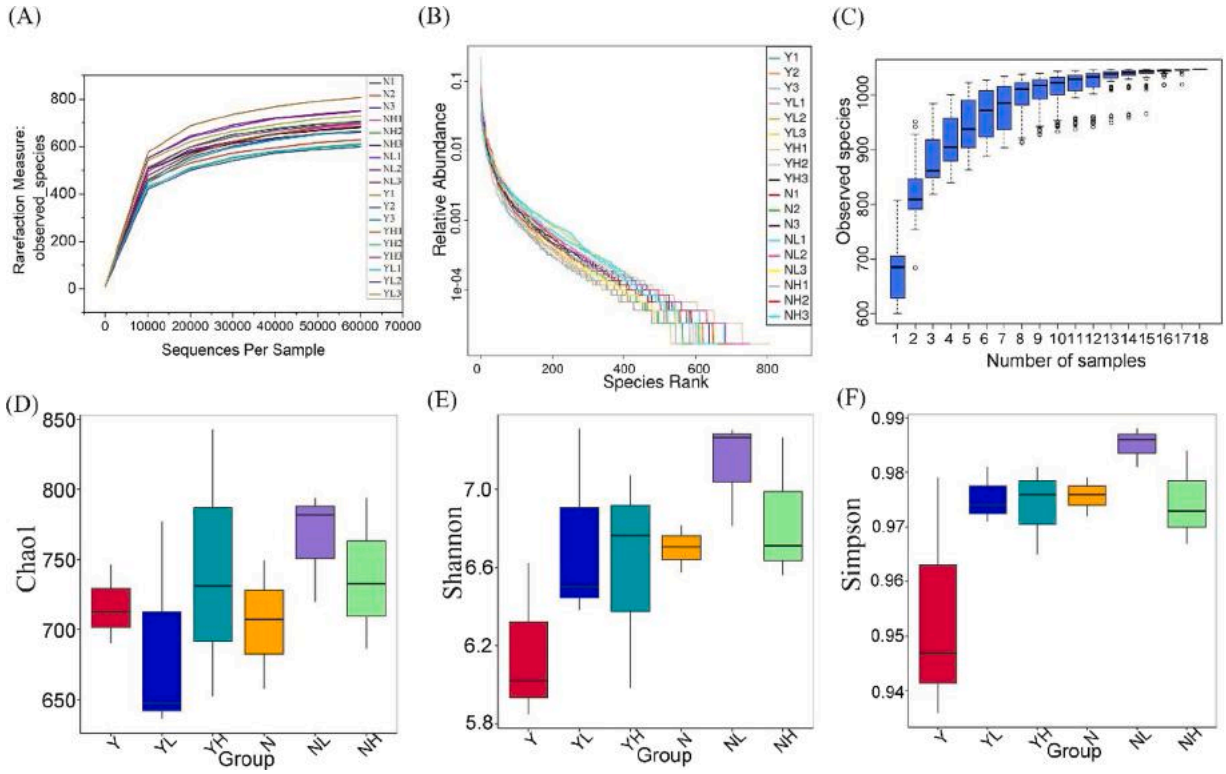

Fig. 1. Effect of HEPs on the richness and diversity of fecal microbiota in HEPs-treated mice. (A) Observed species curve. (B) Rank abundance curve. (C) Specaccum curve. (D) Chao1 index. (E) Shannon index. (F) Simpson index.

3. Results and discussion

3.1. Basic physicochemical properties of HEPs

The crude polysaccharides of *Hericium erinaceus* were obtained by hot water extraction and ethanol precipitation with a yield of 2.735 %. Total carbohydrate content of the crude polysaccharides was determined to be 53.36 % (including 4.43 % reducing sugars), while the uronic acid content was 32.56 %.

3.2. The Effects of HEPs on body weight in mice

A recent study indicated that polysaccharide of *Hericium erinaceus* attenuated colitis in C57BL/6 mice *via* enhancing expression of intestinal inflammatory cytokines, and regulating the composition of the gut microbiota (Ren et al., 2018). Therefore, these mice were given HEPs for 28 consecutive days to estimate the effect of HEPs on modulating the composition of the gut microbiota.

Table 1 shows the effects of HEPs on the body weight of 8-week-old and 6-month-old male mice. The results suggested that the administration of HEPs inhibited the weight gain of adult mice (8-week-old), while did not significantly inhibited the weight gain of middle-aged and old mice (6-month-old). The alteration of body weight could be eventually related to the alteration of the gut microbiota after the treatment of HEPs (Kim, Choi, Oh, Ahn, & Unno, 2020).

3.3. The Effects of HEPs on the structure of the gut microbiota

The human gut microbiota plays a major role in maintaining health. It affects expression of genes involved in digesting carbohydrates, proteins, drugs, and providing nutrients for the body (Arumugam et al., 2011; Pennisi, 2010; Strozzi & Mogna, 2008). Three feces samples were selected from each group randomly for high-throughput sequencing. After selecting the effective reads, a total of 1,440,509 clean reads were obtained from 18 samples with an average of 80 028 ± 2439 reads per sample (n = 3 for each group). A series of statistical alpha diversity analysis indices were accessed to evaluate species diversity for each sample. Rarefaction curves can be used to compare species diversity in samples with different sequencing data bulk, and can also be used to explain whether the sequencing data bulk of the samples is reasonable. Rank-abundance curves can be used to explain two aspects of species abundance and species evenness. The smoother the curve is, the more even the distribution of species is. Specaccum curves are an effective tool for investigating sample species composition and predicting species abundance and are widely used in biodiversity and community surveys to determine the adequacy of sample bulk and estimate species richness. Rarefaction curves (observed species index curve) reached a plateau, and rank abundance curves smoothed (Fig. 1A). The Specaccum curves appeared increased and then maintained stable as the number of samples increased (Fig. 1B). These results indicated that the sequencing data were sufficient to reflect the richness and evenness of the microbial community in each sample. Chao1 index is used to evaluate the number of OTU in samples. The bigger chao1 index is, the bigger the number of OTU is, indicating that there are more species in samples. Shannon and Simpson are used to estimate the microbial diversity of samples. The bigger the value is, the bigger the community diversity is. As shown in Fig. 1C, the species diversity was significantly larger in the YL group and the YH group compared with that in the Y group, and the species richness and diversity were significantly larger in the NL group compared with that in the N group, indicating that HEPs enhanced the richness and diversity of gut bacteria.

Based on the sequencing data, the phylogenetic tree (Fig. S1) in groups at the phylum level was drawn. The *Firmicutes* and *Bacteroidetes* appeared the two dominant phyla in the fecal microbial community of the experimental group and control group of the juvenile and adult mice. There were no differences in the fecal microbial community between the adult mice and middle-aged and old mice at the phylum level. *Firmicutes* and *Bacteroidetes*, as the two most abundant types of bacteria, have been

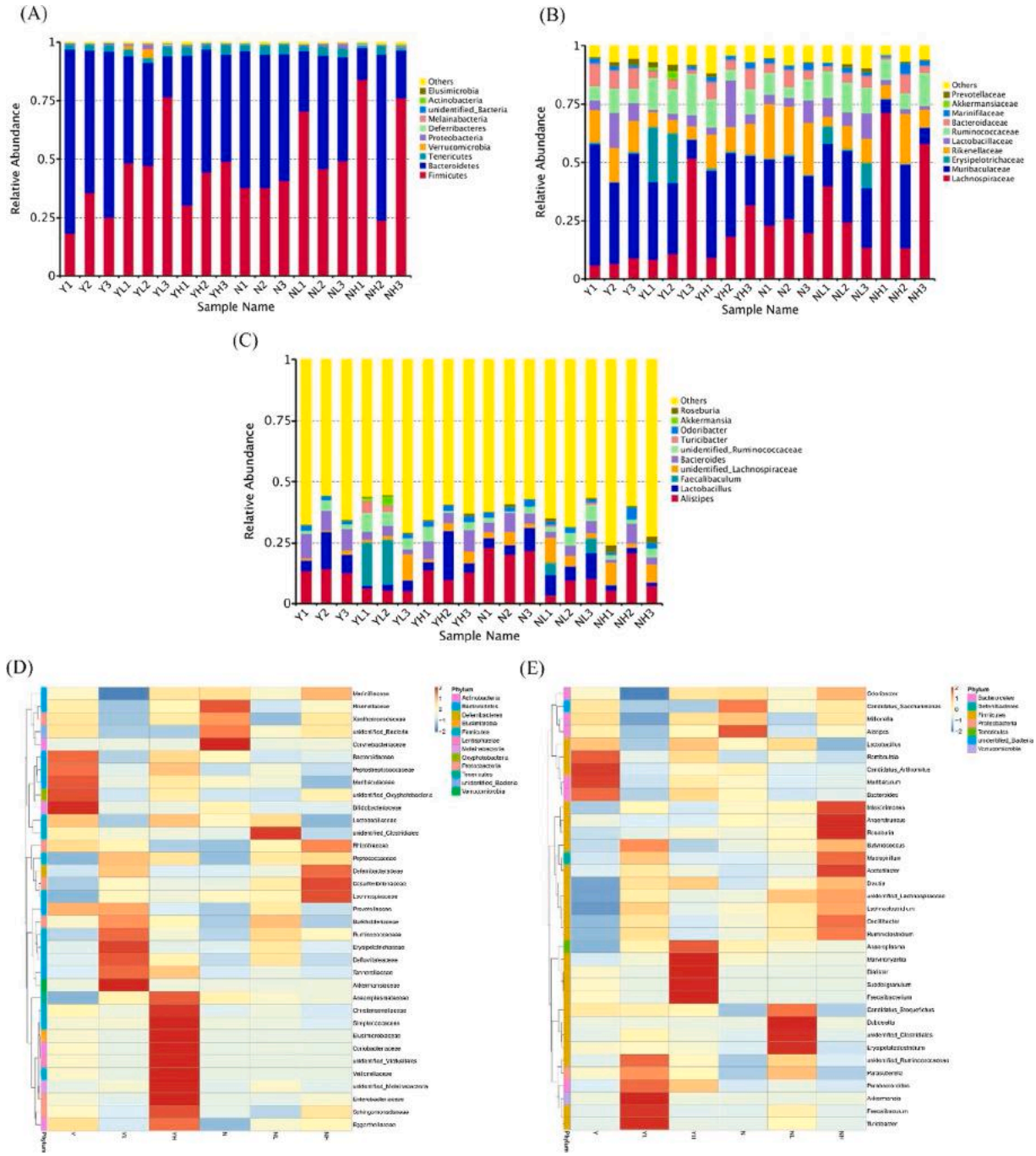

Fig. 2. Effect of HEPs on the gut microbiota structure in HEPs-treated mice. The relative abundance of gut microbiota at the (A) phylum, (B) family and (C) genus levels. The taxonomic heatmap of gut microbiota at the (D) phylum and (E) genus levels.

reported to contain a vast number of glycoside hydrolases, which play an important role in the degradation of non-digestible polysaccharides. Furthermore, *Bacteroidetes* has more carbohydrate metabolic pathways and polysaccharide-degrading enzymes than *Firmicutes* to ferment glycans to short-chain fatty acids (Mahowald et al., 2009; Turnbaugh et al., 2009).

Heatmap can reflect data in a two-dimensional matrix or table with color changes, and it can intuitively represent data values with defined color depth. As shown in Fig. 2A, compared to the control group (the Y group and the N group), HEPs decreased the relative abundance of *Bacteroidetes*. The changes at the family level also appeared in Fig. 2B. HEPs treatment increased the relative abundance of *Lachnospiraceae, Ruminococcaceae* and *Akkermansiaceae*, and decreased the relative abundance of *Muribaculaceae*, *Rikenellaceae*, *Lactobacillaceae* and *Bacteroidaceae* in the experimental group and control group of both adult mice and middle-aged and old mice, which was in accordance with the results shown in Fig. 2D. In addition, *Erysipelotrichaceae*, *Enterobacteriaceae*, *Christensenellaceae* and *Coriobacteriaceae* were increased, while *Bifidobacteriaceae* and *Peptostreptococcaceae* were reduced in the fecal microbial community of the HEPs-treated adult mice (Fig. 2D). The relative abundance of *Rhizobiaceae*, *Desulfovibrionaceae* and *Lachnospiraceae* were increased, while the relative abundance of *Corynebbacteraceae* and *Rikenellaceae* appeared to decrease in the fecal microbial community of the HEPs-treated middle-aged and old mice (Fig. 2D). At the genus level, the genera with significant alteration between the experimental group and the control group also appeared in *Alistipes*,

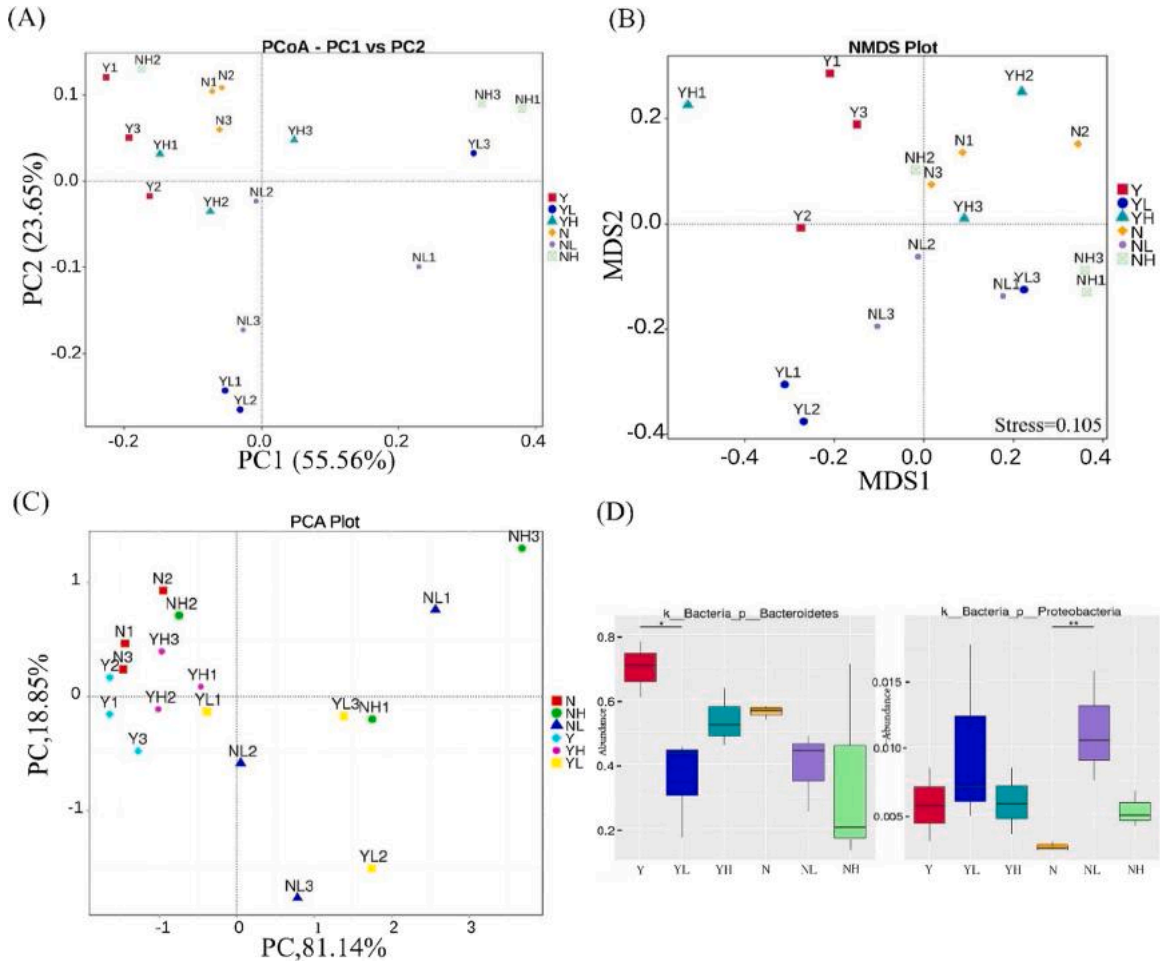

Fig. 3. β-Diversity analysis of gut microbiota. (A) PCoA analysis based on weighted UniFrac distances. (B) NMDS analysis. (C) PCA analysis based on MetaStats analysis and box plot. (*$p < 0.05$ compared with Y group, $n = 3$, **$p < 0.05$ compared with N group, $n = 3$).

Lactobacillus, *Bacteroides* and *Faecalibacterium* (Fig. 2C and E). These results were in agreement with the previous reports[7,8]. At the family level, *Lachnospiraceae* and *Ruminococcaceae,* belonging to the main flora of *Firmicutes*, are the main butyrate-producing bacteria, suggesting that HEPs promote intestinal health possibly by increasing the number of probiotics to promote the synthesis of short-chain fatty acids (Louis & Flint, 2017; Ye et al., 2020).

The overall structural changes of the gut microbiota were analyzed using principal co-ordinates analysis (PCoA; Fig. 3A), and nonmetric multidimensional scaling analysis (NMDS; Fig. 3B). As shown in the PCoA plot, the cumulative variance contribution rate of the first two principal components were 79.21 %. The PCoA plot displayed that the microbial community structure in YL group was significantly different from Y group, and the microbial community structure in NL group was significantly different from N group. Similar observation can also be obtained in NMDS analysis. After MetaStats analysis of the sequencing data, the results were exhibited in the PCA plot (Fig. 3C), which appeared consistent with these in PCoA and NMDS analysis. According to the box plot in Fig. 3C, the significant differences were distributed in *Bacteroidetes* and *Proteobacteria*. In addition, according to the unweighted UniFrac distances calculated between the samples, the heatmap (Fig. 4A) was drawn, revealing that the effect of HEPs on the bacterial composition of adult mice was better than that of middle-aged and old mice.

The specific bacterial taxa among different groups were identified by using the linear discriminant analysis effect size (LEfSe) method (Fig. 4B and C). The results indicated that the microbial community in YL group was distinct from Y group. The YL group was enriched in *Eubacteriaceae* at the family level, *Anaerofustis* at the genus level, *Clostridiales* bacterium CIEAF 020 at species level, while the Y group was enriched in *Bacteroidetes* at the phylum level, *Bacteroidia* at the class level, *Bacteroidales* at the order level, *Candidatus Arthromitus* at the genus level.

3.4. Annotation of the Gut bacterial function

The Tax4fun database, including species (Silva database: Silva123) and functional relationships (Kyoto Encyclopedia of Genes and Genomes, KEGG), is used to annotate the gut bacterial function.

Fig. 5 was drawn based on *t*-test to compare the functional prediction of altered gut microbiota at level II between the Y and YL group, YL and YH group, N and NL group, respectively. It was determined that the membrane transport, glycan biosynthesis and metabolism, transport and catabolism, folding sorting and degradation, xenobiotics biodegradation and metabolism were significantly altered in HEPs-treated mice. In addition, all functional genes were divided at level I, level II, level III and level K to draw the heatmaps (Fig. S2). In the NH group, genes relevant to amino acid metabolism, immune system, cell growth and death, cell motility, metabolism and signal transduction were markedly up-regulated. Genes relevant to cancers, poorly characterized, translation, replication and repair, nucleotide metabolism, lipid metabolism, enzyme families and drug resistance were also up-regulated. In the YL group, multiple pathways appeared to be influenced with three of them evidently down-regulated, involving translation, replication and repair and nucleotide metabolism, while three pathways including endocrine

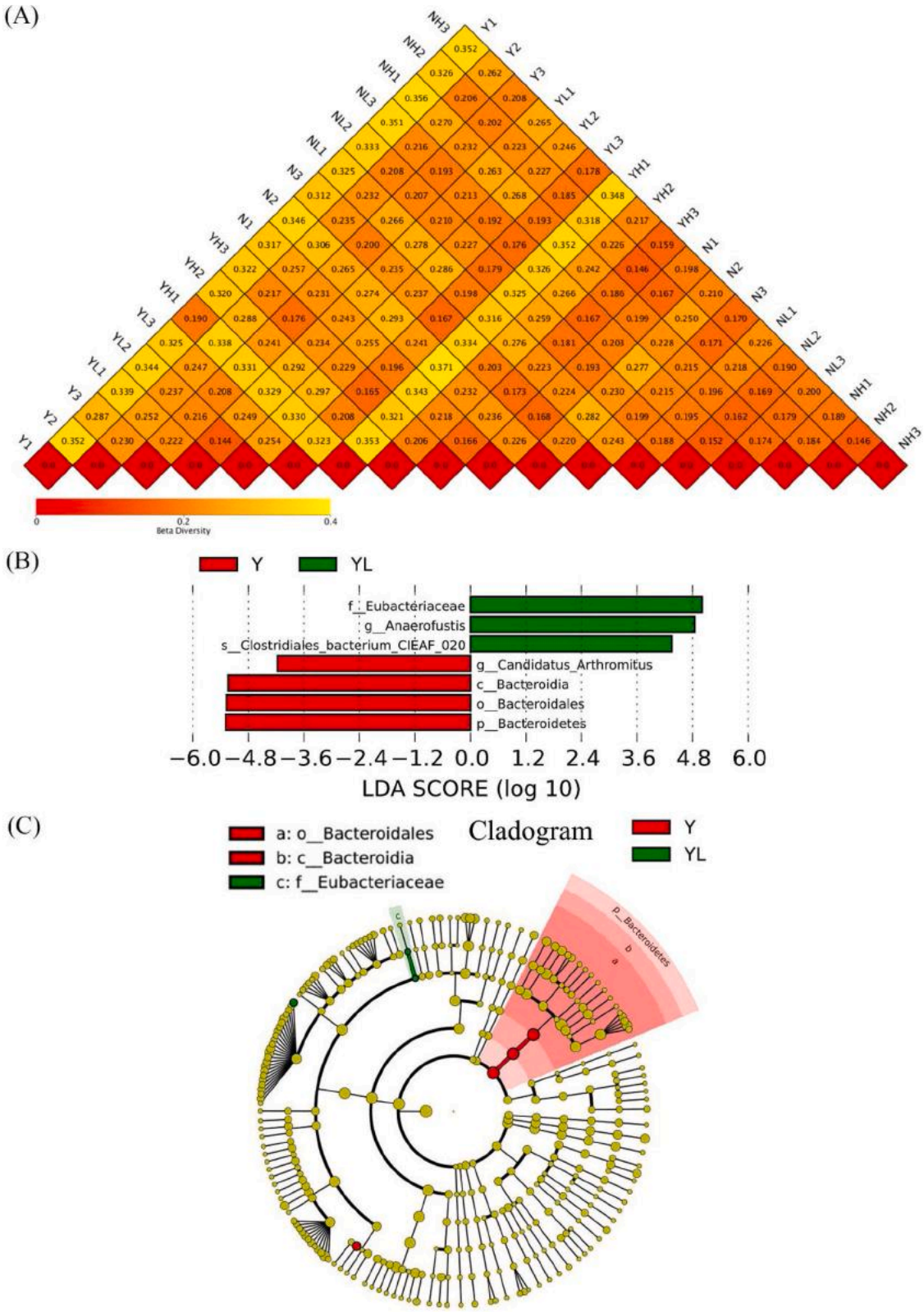

Fig. 4. β-Diversity heatmap and LEfSe analysis results. (A) β-diversity heatmap based on unweighted UniFrac distances. Cladogram showing the phylogenetic relationships of (B) bacteria taxa and (C) LDA scores between the Y group and the YL group.

system, metabolism of cofactors and vitamins, biosynthesis of other secondary metabolites, transport and catabolism were down-regulated. Furthermore, administration of HEPs evidently changed the gut microbiota of both adult mice and middle-aged and old mice, according to the heatmaps of the four levels of functional prediction. Based on the sequencing date, the route diagrams of predicted microbial metabolism of the YL and NH groups were drawn (Fig. S3).

In the recent years, there are studies reporting on the combined

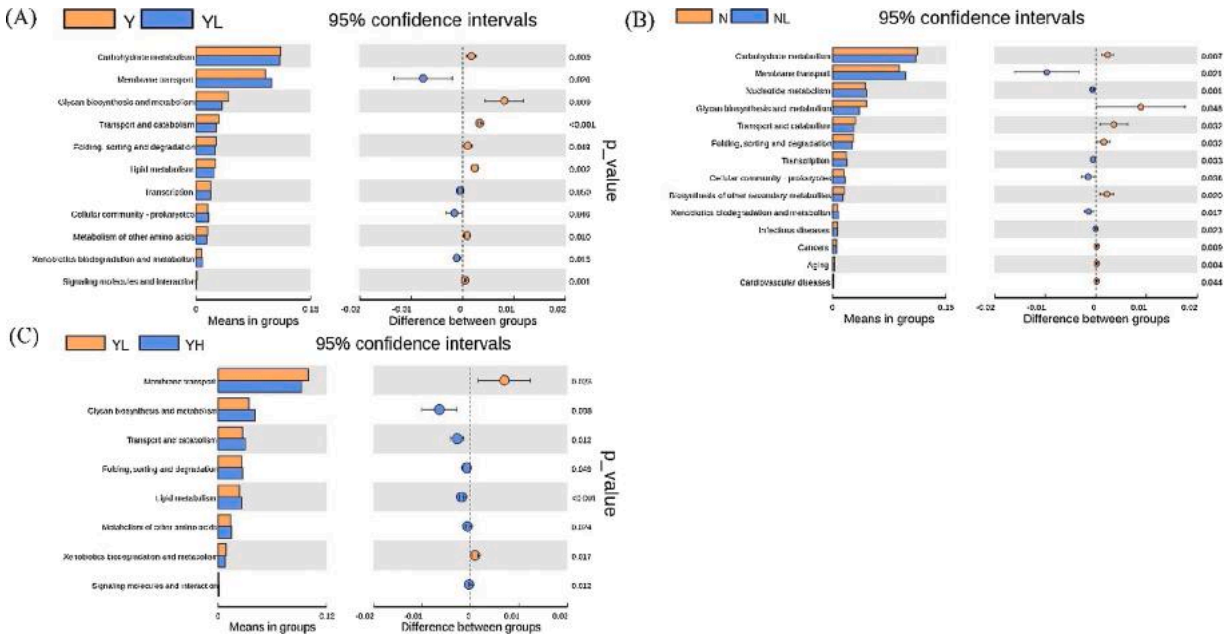

Fig. 5. Functional prediction of altered gut microbiota by Tax4Fun analysis based on KEGG (Kyoto Encyclopedia of Genes and Genomes) pathways at level II in mice. Differences were considered significant at $P < 0.05$.

fermentation of glycans and microbes modulated the gut microbiota and immune responses. Although the enterocyte separates the luminal microbiota from the body's professional immune cells, the invasion of pathogens still can be detected by epithelial cells delivering signals to immune cells *via* the pattern recognition receptors (PRRs) (Abreu, Fukata, & Arditi, 2005; Peterson & Artis, 2014). Whole plant fibre combined with fibre-digesting probiotics could increase immune-modulating activities of the components, and futher to modulate the gut microbiota. The prebiotic plant fibre increased the abundance of *Oscillospira* in caecum, *Akkermansia* in faeces while modulating *Bacteroides* in faecal samples at the genus level (Shinde et al., 2020). The abundance of the genus *Bacteroides* was significantly and negatively correlated with faecal lipopolysaccharide (LPS) levels. Chronic immune cell activation is partly caused by the LPS-mediated stimulation of toll-like receptor 4 in the gut (Yoshida et al., 2020). Therefore, we further examined whether polysaccharides might play an immunoregulatory role in intestinal flora.

3.5. Effect of HEPs on cytotoxicity, nitric oxide and phagocytic activity in RAW264.7 cells

We previously evaluated HEPs fermentation *in vitro* (Yang et al., 2018). In order to understand the process of metabolizing polysaccharides *in vivo*, we isolated bacterial solutions from small intestine (SI), cecum, colon and feces of mice to ferment HEPs *in vitro*.

The CCK-8 assay was used to determine if HEPs fermentation supernatants had cytotoxic effects on RAW 264.7 cells. The results showed that HEPs supernatant fermented by small intestine and cecum had no obvious toxic effect on RAW 264.7 cells, while HEPs supernatant fermented by colon and feces enhanced RAW 264.7 cells' proliferation (Fig. 6A).

Nitric oxide as a major mediator of macrophage prevents the invasion of bacteria and tumor cells. Therefore, we examined whether HEPs fermentation supernatant was able to promote the release of NO (Fig. 6B). With the prolongation of fermentation time, the production of NO gradually ascended and then descended (the highest value peaked after 24 h). Compared to the other groups, the HEPs supernatant fermented by colon and feces respectively for 24 h had the most significant effect on the secretion levels of NO in RAW264.7 cells.

Phagocytic activity of macrophages is another major indicator reflecting the functional activation of macrophages. According to Fig. 6C, HEPs supernatants fermented by colon for 24 h and 36 h, and fermented by feces for 24 h significantly stimulated phagocytic activity of macrophages, revealing eventually that the fermentation process could partly increase the functional activation of macrophages.

3.6. Effect of HEPs on the production of cytokines in RAW264.7/Caco-2 cells model

Mucosal tissues as the first defense barrier of the body contain intestinal epithelial cells and immune cells, which are exposed to a large number of antigens. Caco-2 cells are human colon cancer cells that resemble differentiated small intestinal epithelial cells in structure and function and are often used to construct an intestinal model *in vitro* (Fig. 6D). To estimate whether the fermentation supernatant trigger immune responses in macrophages, ELISA assays were used to examine the expression of immune cytokines in the lower compartment. The results showed that the supernatant fermented by colon and feces stimulated the expression of cytokines IL-6, IL-10, IFN-γ and TNF-α in supernatants (Fig. 6E–H), speculating that the fermentation process of colon and feces could promote immune responses in macrophages. We deduced that the better immuno-stimulating effect of the colon and feces group may be related to the involvement of more bacteria in the colon and feces in the fermentation. The bacterial density of colon and feces is approximately 10^{10-12}, which is more than that of small intestine and cecum (Hornung, Santos, Smidt, & Schaap, 2018). That may be why the HEPs supernatants fermented by colon and feces had the more significant effect on the release of cytokines than that of small intestine and cecum.

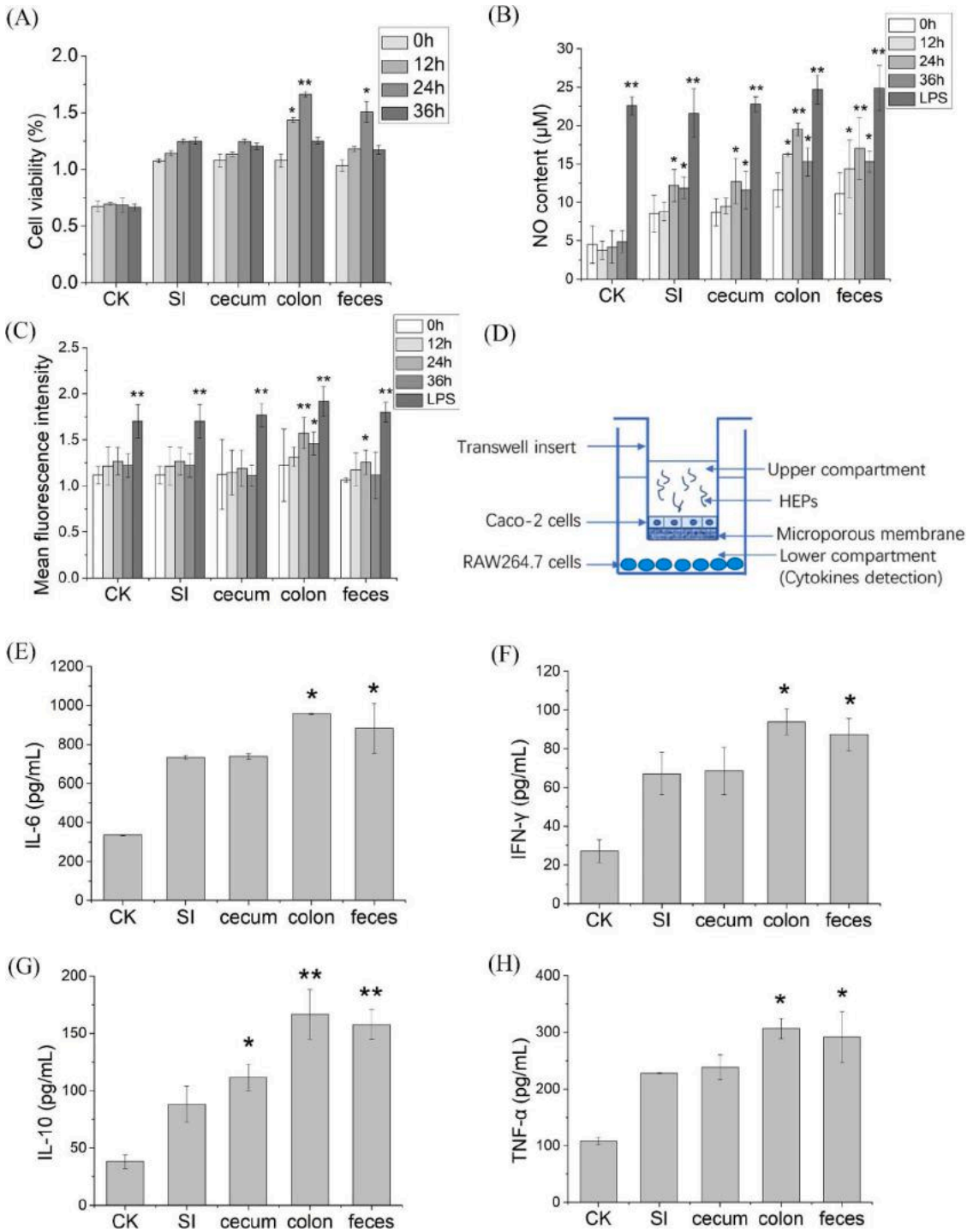

Fig. 6. Effect of HEPs fermented by SI, cecum, colon and feces in RAW264.7 cells. (A) Cell viability. (B) NO production. (C) Phagocytic activity. (D) The co-culture model of RAW264.7/Caco-2 cells. (E) IL-6. (F) IFN-γ. (G) IL-10. (H) TNF-α. Data are presented as mean ± SD (n = 3). *$p < 0.05$, **$p < 0.01$ compared with the CK group. Abbreviation: SI, small intestine.

3.7. Effect of HEPs on the expression of NF-κB, MAPK and PI3K/Akt pathways

The NF-κB, composed of two kinds of Rel family protein, is a set of important transcription factors. As the focal point of the multiple signaling pathways, it plays an important role in the human immune response and cell growth regulation. When inactivated, most of NF-κB and inhibiting protein IκB-α form a complex in the cytoplasm with no transcription activity, while when activated, IκB-α is acidulated and degraded by the ubiquitination pathway. When NF-κB is dissociated, its nuclear localization sequence is exposed and transported to the nucleus to promote the expression of target gene(s) (Beinke & Ley, 2004).

In order to detect whether the HEPs fermentation supernatants activate NF-κB pathway of macrophages, we used Western blot to detect the nucleus displacement of NF-κB p65 (Fig. 7A). The quantified expression of IκB-α and p65 is shown in the histogram (Fig. 7D and E). After the treatment of HEPs supernatants, the content of IκB-α and p65 in cytoplasm decreased in the most groups, while that in the group of colon and feces appeared to increase. In addition, in the nucleus p65 in the group of colon and feces increased. These results suggested that to a

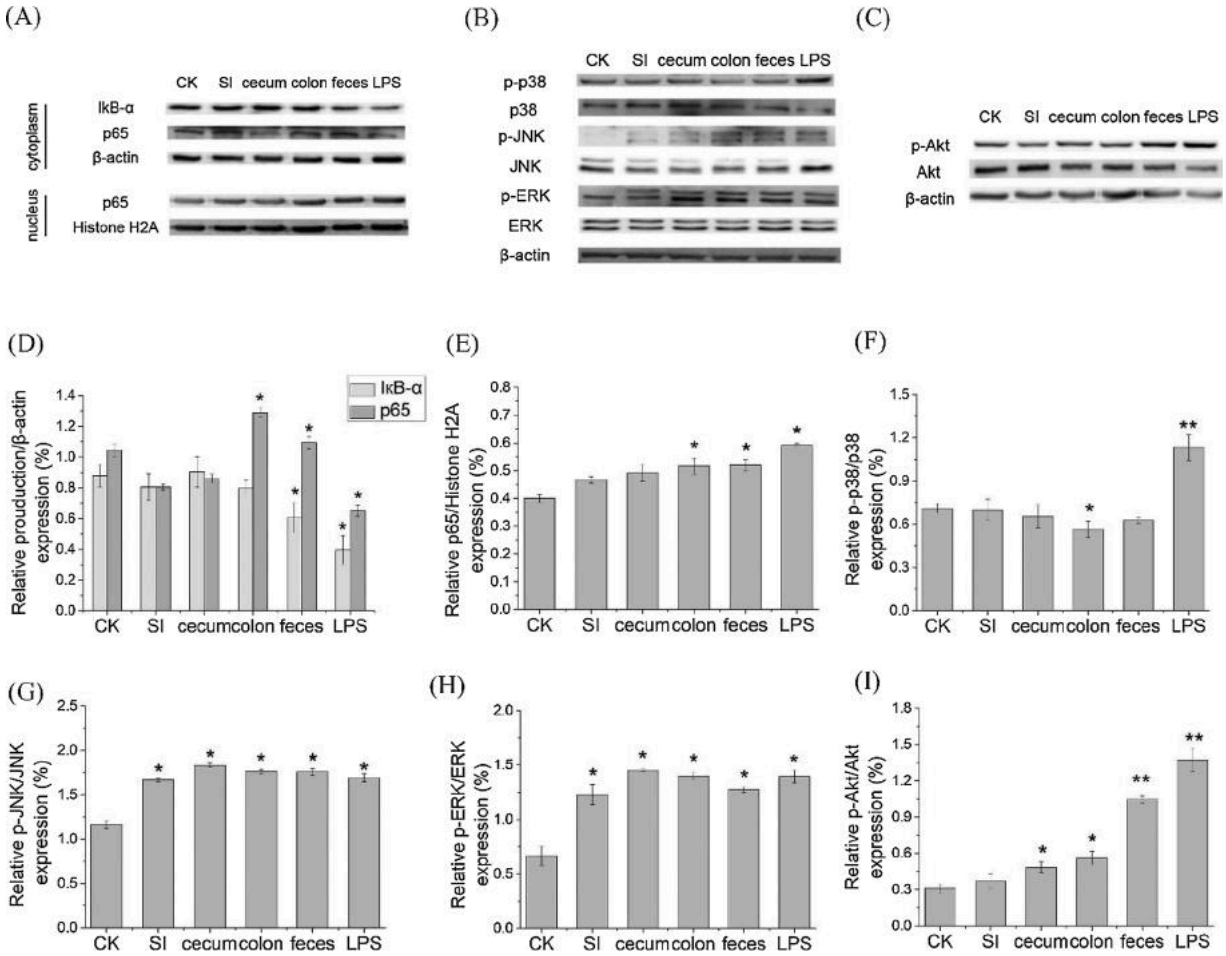

Fig. 7. Immunoregulatory activities of HEP fermented by SI, cecum, colon and feces detected by RAW264.7/Caco-2 cells model. Effect of HEP fermented by SI, cecum, colon and feces on cytoplasm/nuclear (A) NF-κB (p65), (B) MAPK, (C) PI3K/Akt signaling pathways. The quantified expression of (D) IκB-α or p65/β-action on cytoplasm, and (E) p65/Histone H2A on nuclear. The quantified expression of (F) p-P38/P38, (G) p-JNK/JNK, (H) p-ERK/ERK. The quantified expression of (I) p-Akt/Akt. Data are presented as mean ± SD (n = 3). *p < 0.05, **p < 0.01 compared with the CK group. Abbreviation: SI, small intestine.

certain degree HEPs supernatants, fermented by colon and feces, induced the degradation of IκB-α and hence nuclear translocation of NF-κB p65 in RAW264.7 cells, and thus improved the transcriptional activity of NF-κB. We can deduce that HEPs supernatants fermented by colon and feces activate the NF-κB signaling pathway in macrophages.

The MAPK and PI3K/Akt pathways are important pathways to transmit extracellular signals to cells and participate in regulating the release of NO and TNF-α induced by polysaccharide stimulation. Therefore, we further investigated the role of three parallel signaling pathways in MAPK pathway (p38, JNK, ERK) and Akt signaling pathway in the process of the HEPs fermentation supernatants activating macrophages. The results showed that after a stimulation of the four HEPs fermentation supernatants, the phosphorylation of JNK and ERK proteins in MAPK pathway was significantly upregulated (Fig. 7B and F–H). Moreover, the phosphorylation of Akt protein in the PI3K/Akt pathway significantly upregulated after the stimulation of HEPs supernatants fermented by colon and feces (Fig. 7C and I). These results revealed that HEPs supernatants fermented by colon and feces could enhance the transcriptional activity of JNK, ERK and Akt. The MAPK pathway and PI3K/Akt pathway may be involved in the immunoregulation of HEPs supernatants fermented by colon and feces. It was reported that the hydrolyzed derivatives of *Cetraria islandica* polysaccharides could activate RAW264.7 cells through production of NO and mRNA expression of iNOS, IL-1, TNF-α, IL-6, IL-10 and IL-12 through the NF-κB and MAPK signaling pathways (Surayot et al., 2019). Previous reports (Ghosh, Khatua, & Acharya, 2019; Ren, Zhang, & Zhang, 2020) are consistent with the present results.

3.8. The role of NF-KB, MAPK and PI3K/Akt pathways in HEPs fermentation supernatants activation of macrophages

To further verify whether HEPs fermentation supernatants are involved in the stimulation of RAW264.7 cells to secrete NO and TNF-α in NF-κB, MAPK, and PI3K/Akt signal transduction pathways, we used specific inhibitors, such as NF-κB pathway inhibitors (BAY11-7082), MAPK pathway inhibitors (SB203580, SP600125, PD98059) and Akt pathway inhibitors (LY294002) to block the respective signaling pathways. The RAW264.7 cells were pretreated by inhibitors for 1 h and stimulated by HEPs fermentation supernatants for 24 h. The content of NO and TNF-α in the lower compartment of RAW264.7/Caco-2 cells model was determined respectively. The secretion levels of NO and TNF-αwere shown in the histogram (Fig. 8). The NF-κB inhibitor BAY11-7082 almost completely inhibited the secretion of NO and TNF-α in the group of HEP supernatants fermented by colon and feces respectively, while p-p38 inhibitor SB203580 had no effect on the secretion of NO and TNF-α. These results indicated that NF-κB pathway eventually takes part in regulation of the immune responses induced by HEPs fermentation supernatants.

We further examined the protein expression of IκB and p65 in the NF-κB pathway, p38, JNK and ERK in the MAPK pathway, and Akt in the PI3K/Akt pathway by using Western blot (Fig. 9A–E). The quantified expression of these proteins is shown in the histogram (Fig. 9F and J).

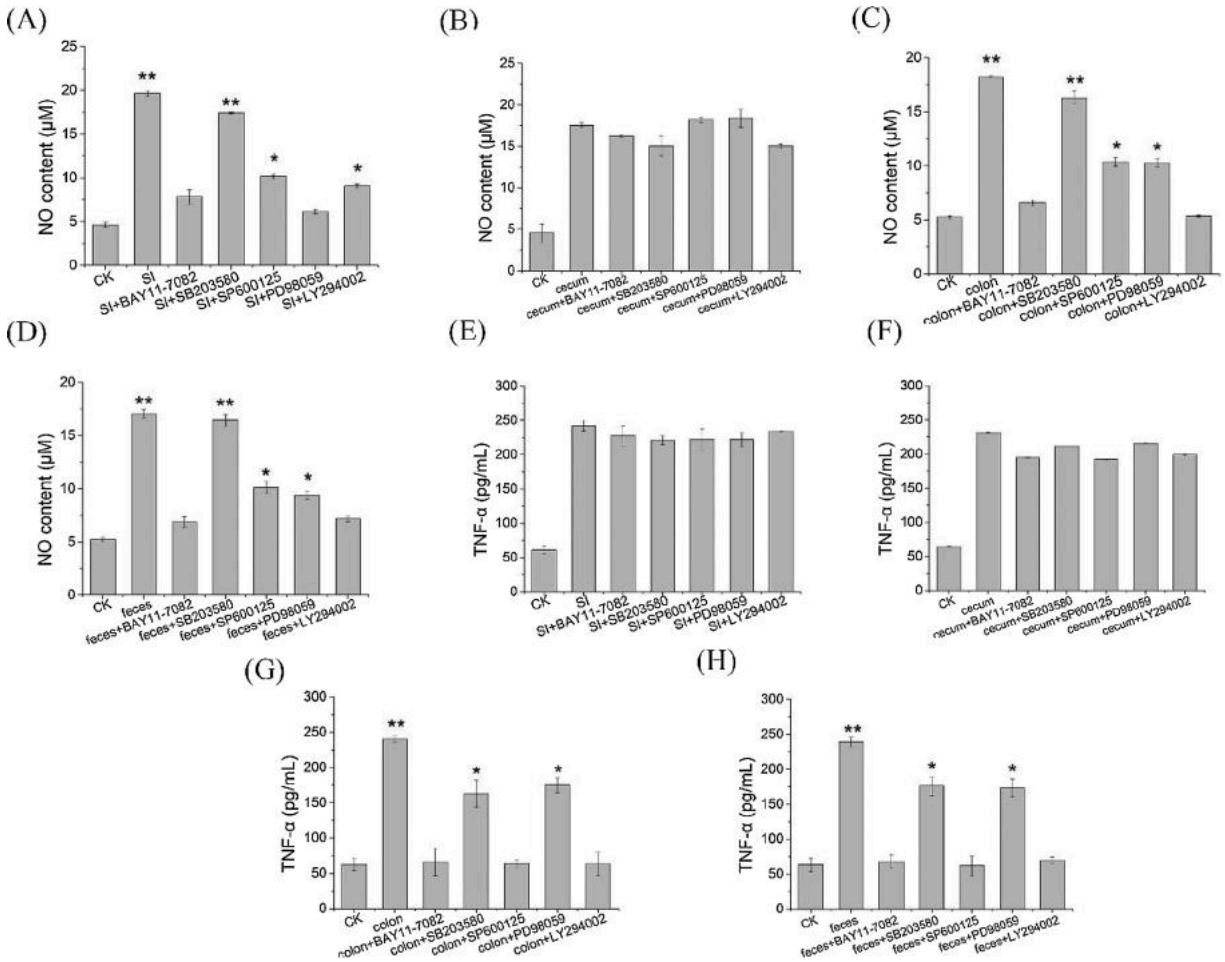

Fig. 8. Effect of inhibitors of NF-κB, MAPK and PI3K/Akt signaling pathways on the release of NO and TNF-α in RAW264.7/Caco-2 cells model. Inhibitory effects of BAY11-7082, SB203580, SP600126, PD98095 and LY294002 on the release of (A) to (D) NO and (E) to (H) TNF-α. Data are presented as mean ± SD (n = 3). *$p < 0.05$, **$p < 0.01$ compared with the CK group. Abbreviation: SI, small intestine.

The five pathway inhibitors reduced the supernatant-induced phosphorylation of related proteins to varying degrees. Therefore, all three pathways are involved in HEPs supernatant-induced immunomodulatory process.

Low molecular weight extracts from fermented sourdough could promote the phosphorylation status of NF-κB expressed as the ratio between the phosphorylated and total form (p-NF-κB/NF-κB), and IκB-α (Gupta et al., 2017; Galli et al., 2018). The activated p-NF-κB/NF-κB is important for the transcription of genes involved in the regulation of the host's immune and inflammatory response (Milanovic, Kracht, & Schmitz, 2014). On the contrary, the expression of IκB-α and NF-κB enhanced the release of anti-inflammatory signals (iNOS and COX2) (Gupta et al., 2017; Park et al., 2016). *Laetiporus sulphureus*-fermented wheat bran (LS) enhanced the intestinal microflora and inflammation status. The LS supplementation tended to have higher ileal and fecal lactic acid bacteria counts and lower ileal coliform counts. The proinflammatory cytokines TNF-α and IL-6 concentration in serum were significantly decreased compared with the wheat bran group (Lin & Lee, 2020). It can be seen that HEPs regulate the immune responses possibly through influencing gut microbiota. As HEPs performed a good ability of regulating the gut microbiota and immune responses, it can be as ingredients to be added into some commercial products, for example, beverages, yogurt, tea, pharmaceuticals and so on. There are few products with high-added value in market, such as health care products, cosmeceuticals and fungal drugs, so *Hericium erinaceus* polysaccharides have great commercial development potential.

4. Conclusions

To sum up, the oral administration of HEPs could change the diversity and the abundance of gut microbiota. The alteration of flora further leads to the change of the proportion of probiotics, finally modulating the immune responses. The HEPs could regulate the immune function *via* NF-κB, MAPK and PI3K/Akt signaling pathways. The present study provided a valuable resource for investigating processes, functions and pathways of effect of HEPs on immunomodulatory effects and gut microbiota.

CRediT authorship contribution statement

Yang Yang: Conceptualization, Data curation, Formal analysis, Methodology, Visualization, Writing - original draft. **Haiqing Ye:** Conceptualization, Methodology. **Changhui Zhao:** Conceptualization, Methodology. **Li Ren:** Investigation, Methodology, Software. **Cuina Wang:** Investigation, Methodology, Supervision. **Milen I. Georgiev:** Validation, Writing - review & editing. **Jianbo Xiao:** Validation, Writing - review & editing. **Tiehua Zhang:** Funding acquisition, Project administration, Resources, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

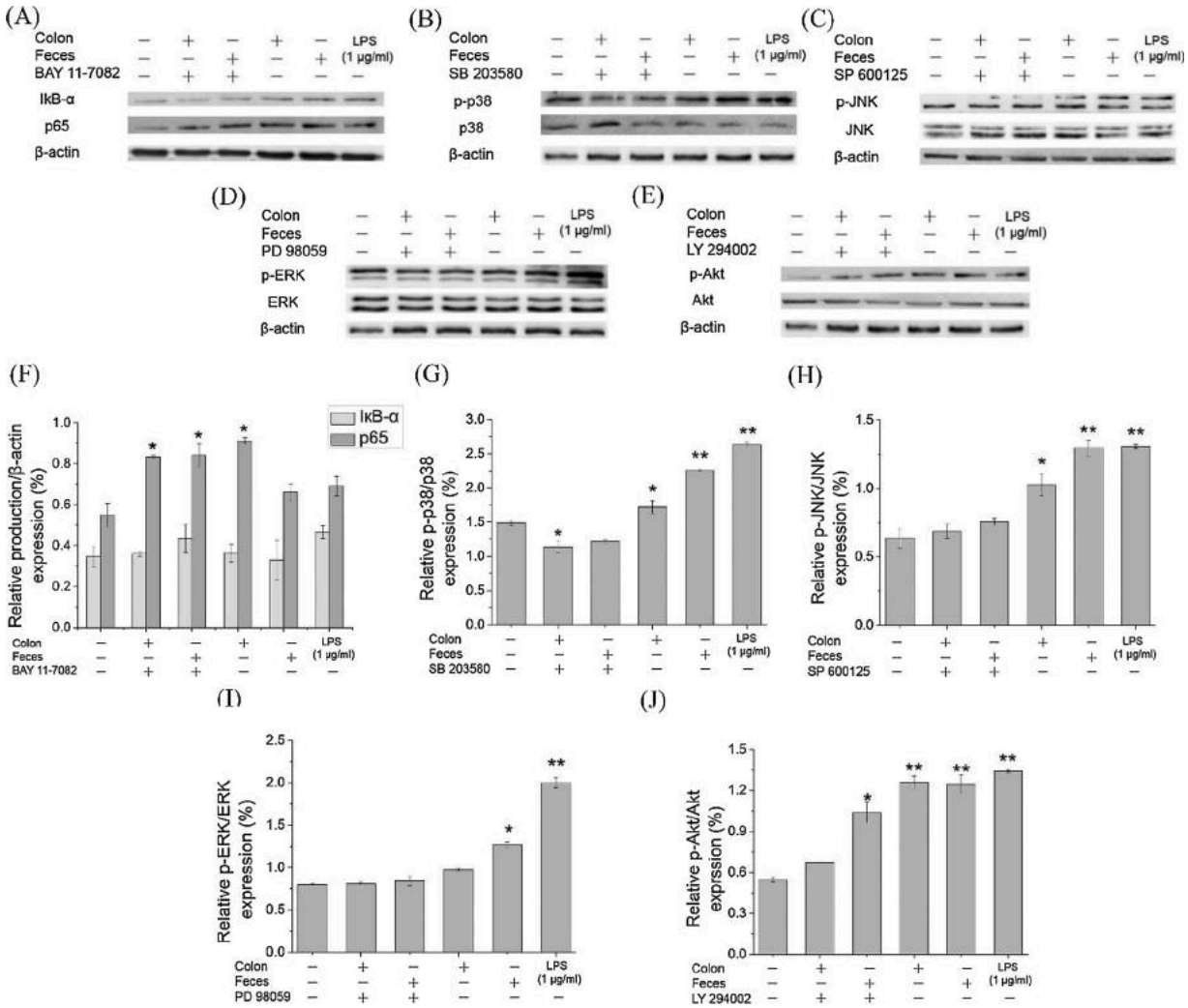

Fig. 9. Effect of inhibitors of NF-κB, MAPK and PI3K/Akt signaling pathways on the phosphorylation of related proteins in RAW264.7/Caco-2 cells model. Inhibitory effects of (A) BAY11-7082, (B) SB203580, (C) SP600126, (D) PD98095 and (E) LY294002 on the NF-κB, MAPK and PI3K/Akt signaling pathways, and the quantified expression of (F) IκB-α or p65/β-action, (G) p-P38/P38, (H) p-JNK/JNK, (I) p-ERK/ERK and (J) p-Akt/Akt. Data are presented as mean ± SD (n = 3). *p < 0.05, **p < 0.01 compared with the CK group. Abbreviation: SI, small intestine.

Acknowledgments

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.carbpol.2021.117668.

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